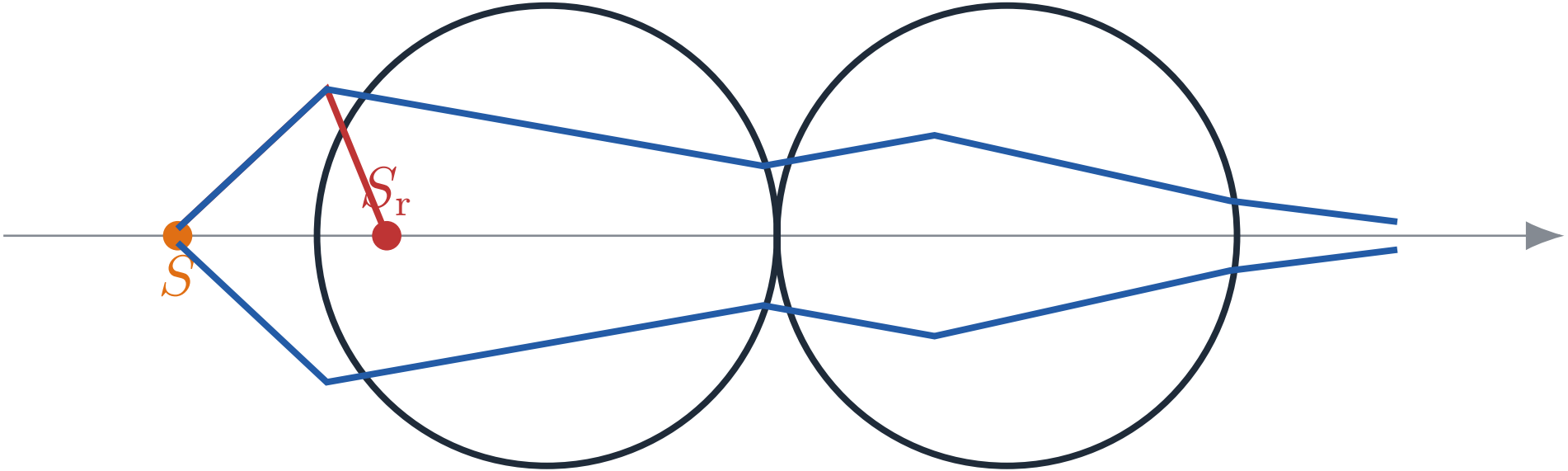

$$\Phi_{\text{bubble}} = -\frac{2(n-1)}{R}, \qquad \text{像重合条件给出 } n \approx 1.3$$